MICROSOFT HACKATHON · THEME: KNOWLEDGE PRESERVATION

Azure Recall

Knowledge that outlives the project.
Centralized, owner-governed knowledge for teams and their AI agents, built on Azure.

Document	Solution documentation with architecture and workflow diagrams
Status	Working prototype deployed on Azure (App Service, Entra ID, Cosmos DB, Azure OpenAI, Application Insights)
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1. Executive summary

Teams now solve most problems with AI assistants such as Claude, ChatGPT and GitHub Copilot. The investigation behind each solution (design choices, failed attempts, the settings that mattered) stays in one team's chat history. When another client brings the same problem, a new team starts from zero.

Azure Recall is a central knowledge layer on Azure. It **captures** lessons from AI sessions and postmortems, **governs** them through projects with accountable owners and Microsoft Entra ID, and **returns** them at the moment a person or coding agent starts similar work. Restricted knowledge can be *discovered* (title and owner) without being *exposed*; reading it requires the owner's approval.

> Offered through Azure Marketplace, Azure Recall creates a dual revenue stream for Microsoft: marketplace transactions and Azure consumption.

2. Problem

Research lives in AI chats
The code is kept; the reasoning, dead ends and fixes stay in a private conversation.

Every new team starts from zero
The same client problem is researched again, with the same mistakes and the same cost.

Written knowledge is not trusted or found
Wikis go stale, there is no owner to ask, and nobody knows whether advice still holds.

Client work is confidential
"Share everything" is not an option, so knowledge stays siloed by default.

3. Solution and approach

Azure Recall sits next to the AI tools teams already use and closes the loop between projects:

1. **Capture.** Paste an AI session, postmortem or PR description, or let the coding agent call `capture_notes`. Azure OpenAI extracts short, reusable lessons and drops secrets and personal data.
2. **Discover.** Describe the new task in plain language. Get an ordered checklist grounded in the organization's own lessons, the relevant entries, coverage gaps, and restricted projects that already solved something similar, with their owners.
3. **Request and approve.** One click sends an access request with a business justification. The project owner approves or denies; approval is bound to the requester's Entra ID identity.
4. **Reuse and improve.** Export the plan as `AGENTS.md` for the coding agent. Votes mark lessons helpful or out of date; stale advice is flagged.

4. Key features

Feature	What it does
Project-based governance	Every project has an owner, client, business domain and visibility: **org-visible** (whole tenant reads) or **restricted** (discoverable, owner-approved reading).
Owner approval workflow	Access requests with justification; only the owner decides; only members write. Identity from Microsoft Entra ID.
AI-assisted capture	Unstructured AI transcripts become structured, tagged lessons in the right project, with redaction of secrets and personal data.
Task-aware discovery	Azure OpenAI ranks lessons for the task and writes a checklist that cites its sources, using only what the user may read.
Gap detection	Shows which areas (security, cost, testing, operations, steps) have no recorded knowledge yet.
Agent integration (MCP)	GitHub Copilot, Claude Code and Cursor connect to `/mcp`: `plan_build`, `search_knowledge`, `capture_notes`, `add_knowledge`, `rate_knowledge`, `request_project_access`. Agents see exactly what their user can see.
Freshness	Helpful / out-of-date votes; age and votes flag stale advice; edits reset the flag.
Reusable infrastructure	Scan Azure resource groups into entries carrying redeployable ARM templates.
Any business domain	Engineering, sales (RFP responses), operations (onboarding playbooks), finance, HR: same model.

5. Architecture

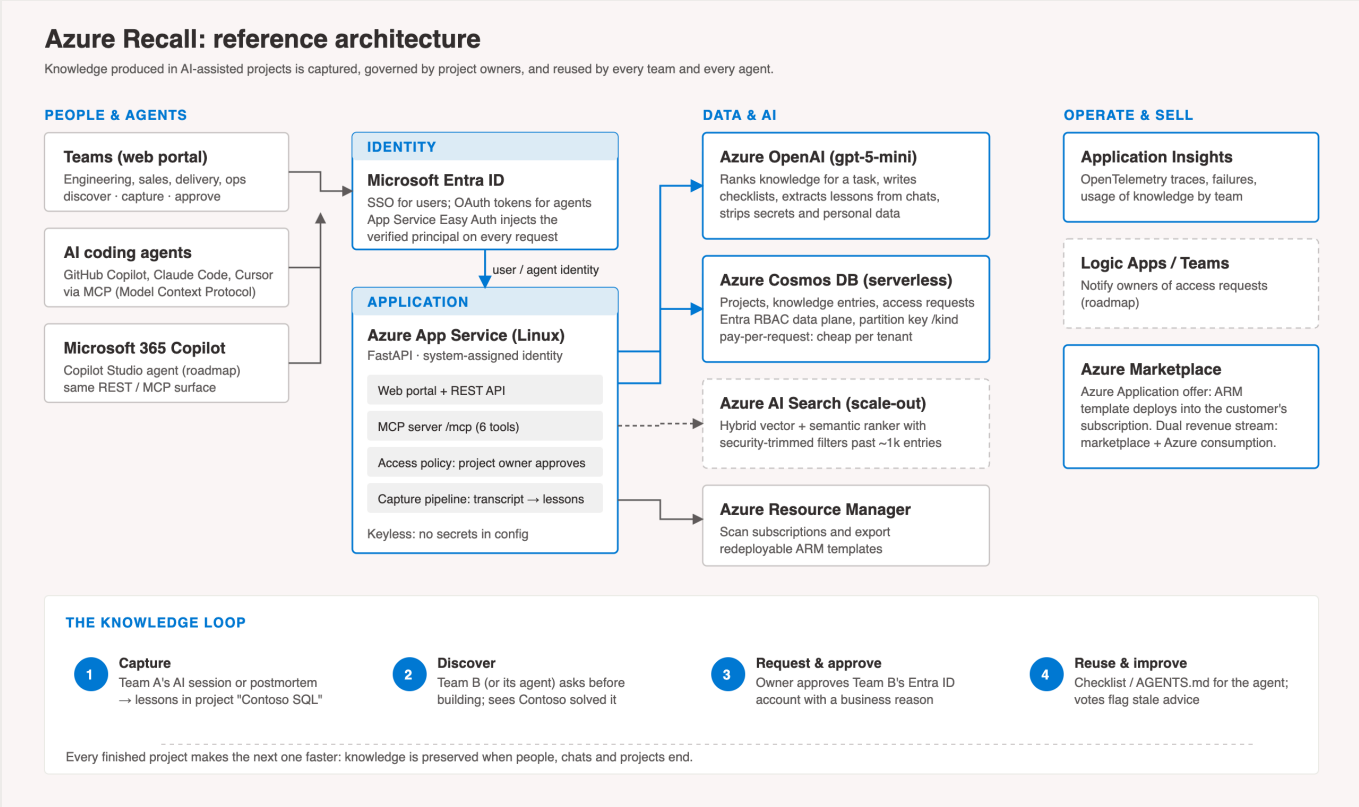


Figure 1. Reference architecture. Dashed boxes are on the roadmap.

Azure / Microsoft service	Role in the solution
Azure App Service (Linux)	Hosts the web portal, REST API and MCP endpoint; stateless, scales horizontally
Microsoft Entra ID (Easy Auth)	SSO for people, OAuth tokens for agents; verified identity drives every access decision
Azure Cosmos DB (serverless)	Projects, knowledge entries, access requests; Entra RBAC on the data plane; pay per request
Azure OpenAI (gpt-5-mini)	Ranking, checklist generation, lesson extraction and redaction
Managed identity + Azure RBAC	Keyless access to Cosmos DB and Azure OpenAI; no secrets in configuration
Application Insights	OpenTelemetry traces, failures, adoption by team
Azure Resource Manager	Exports resource groups as reusable templates
Azure Marketplace	Azure Application offer; ARM template deploys into the customer's subscription

6. End-to-end workflow

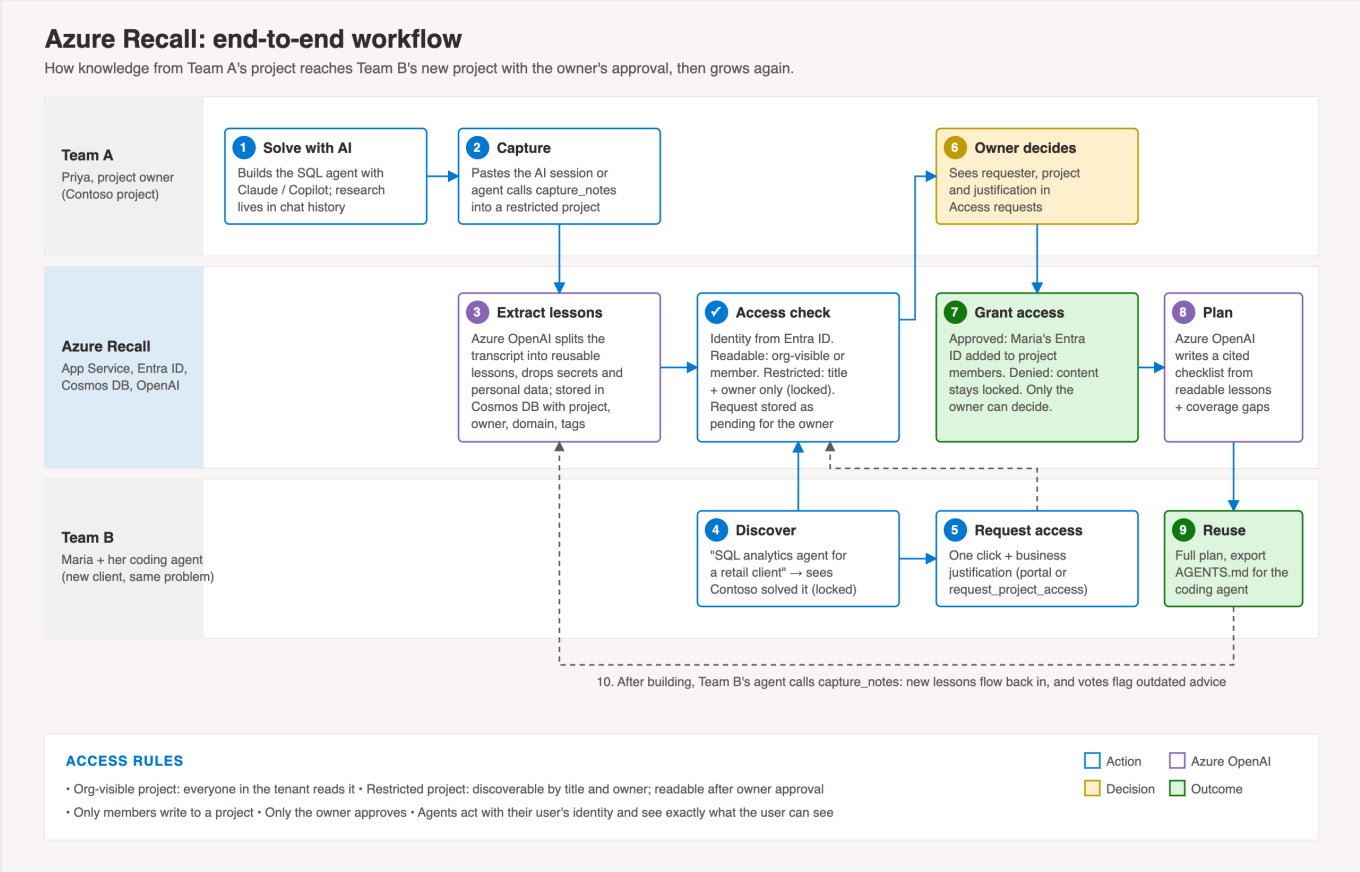


Figure 2. From Team A's AI session to Team B's reuse, with owner approval in between.

Access request sequence

#	Actor	Interaction
1	Maria / her agent	`plan_build("SQL analytics agent for a retail client")`
2	Azure Recall	Returns org-visible lessons + locked project "Contoso Retail SQL Agent" (owner: Priya)
3	Maria	Requests access with justification (portal or `request_project_access`)
4	Priya (owner)	Sees the request in Access requests and approves
5	Azure Recall	Adds Maria's Entra ID account to project members
6	Maria / her agent	Same query now returns the full cited checklist and all lessons; exports `AGENTS.md`

7. Data and access model

One Cosmos DB container `items`, partition key `/kind`:

kind	Fields
project	id, name, owner (UPN), members[], visibility (org / restricted), domain, client, description
entry	id, title, body, tags[], project, domain, author, source, updated, helpful, outdated, template?
request	id, project, owner, requester, reason, status (pending / approved / denied), created, decided

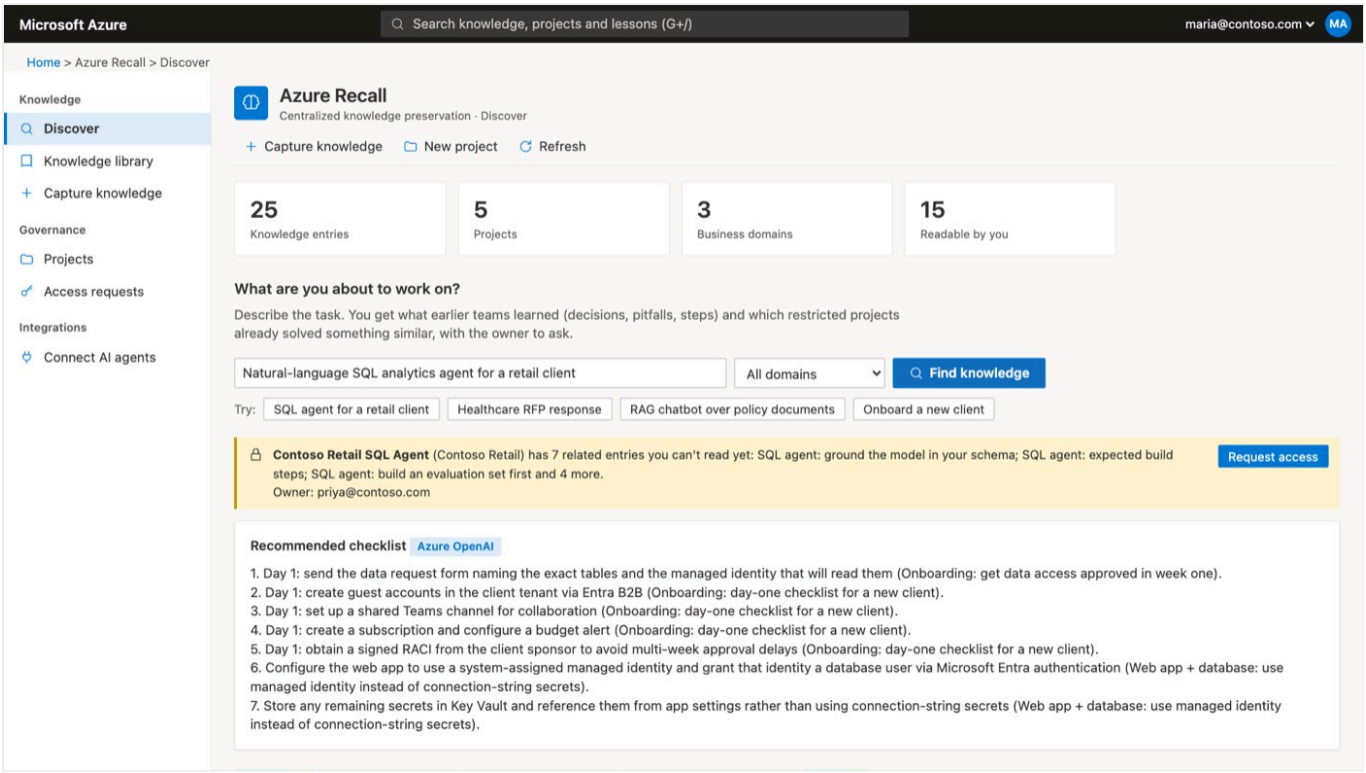
Action	Allowed for
Discover a title in a restricted project	Everyone signed in
Read an entry	Org-visible project, org-wide entry, or project owner / member
Write to a project	Owner / members
Approve or deny access	Project owner only
Delete an entry	Author or project member

8. Security and responsible AI

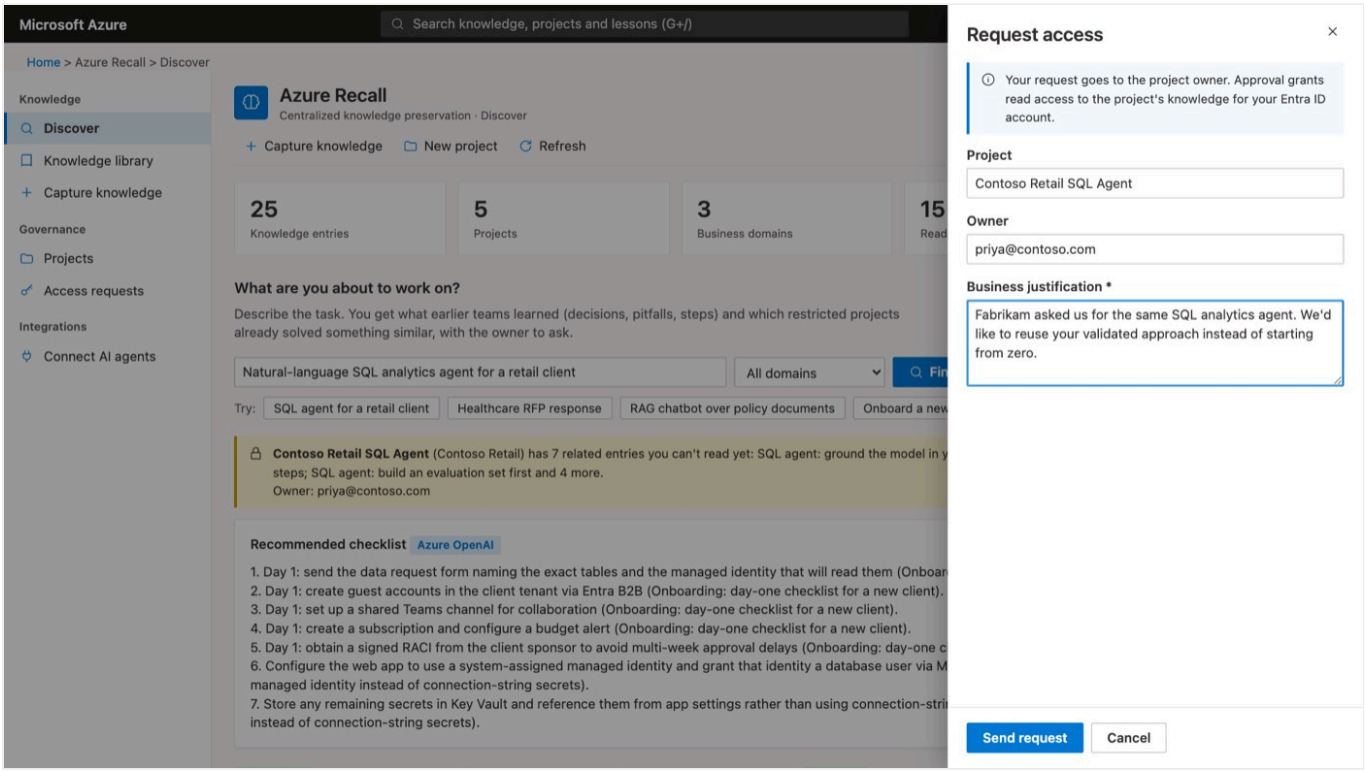
- **Identity:** Microsoft Entra ID via App Service Easy Auth; client-sent identity headers are stripped by the platform.
- **Least exposure:** locked results carry title, project and owner only; content never leaves the API without membership.
- **Keyless:** system-assigned managed identity with Cosmos DB data-plane RBAC and the *Cognitive Services OpenAI User* role.
- **Grounded AI:** checklists use only lessons the user may read and cite them by title; no answers from outside the knowledge base.
- **Redaction:** capture instructs the model to drop secrets and personal data; humans can edit or delete any entry.
- **Data residency:** deployed into the customer's own subscription and region via Marketplace.

9. Product walkthrough

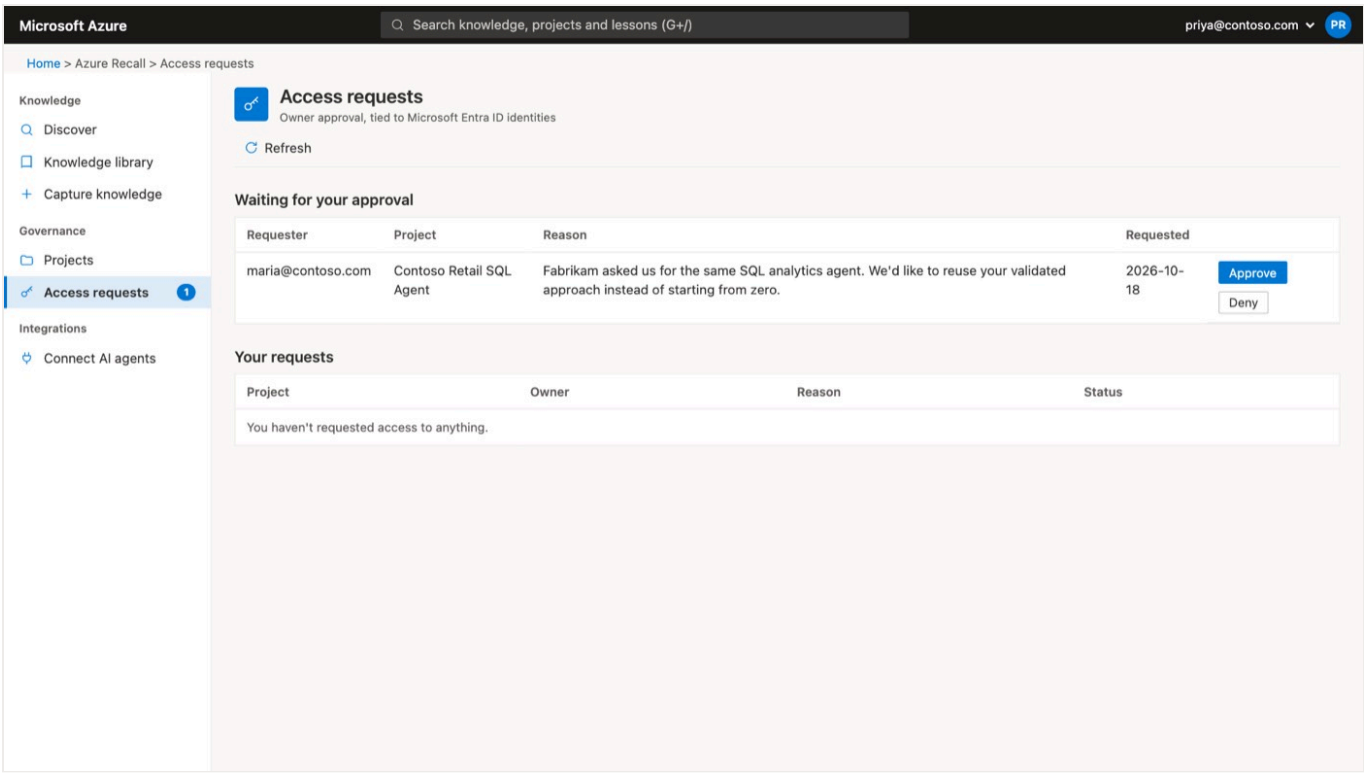
Screenshots from the prototype running on Azure App Service.



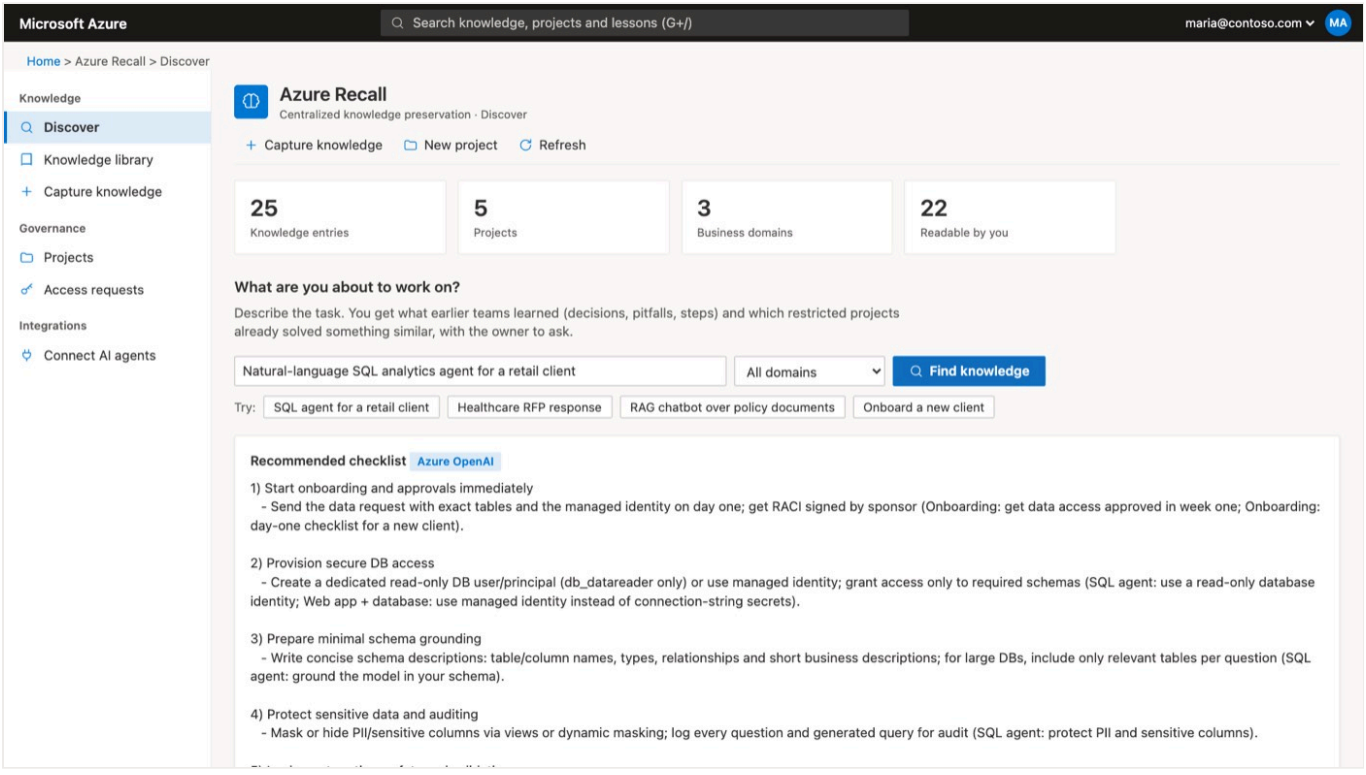
Discover: Maria learns that Contoso's restricted project already solved her problem, and who owns it.



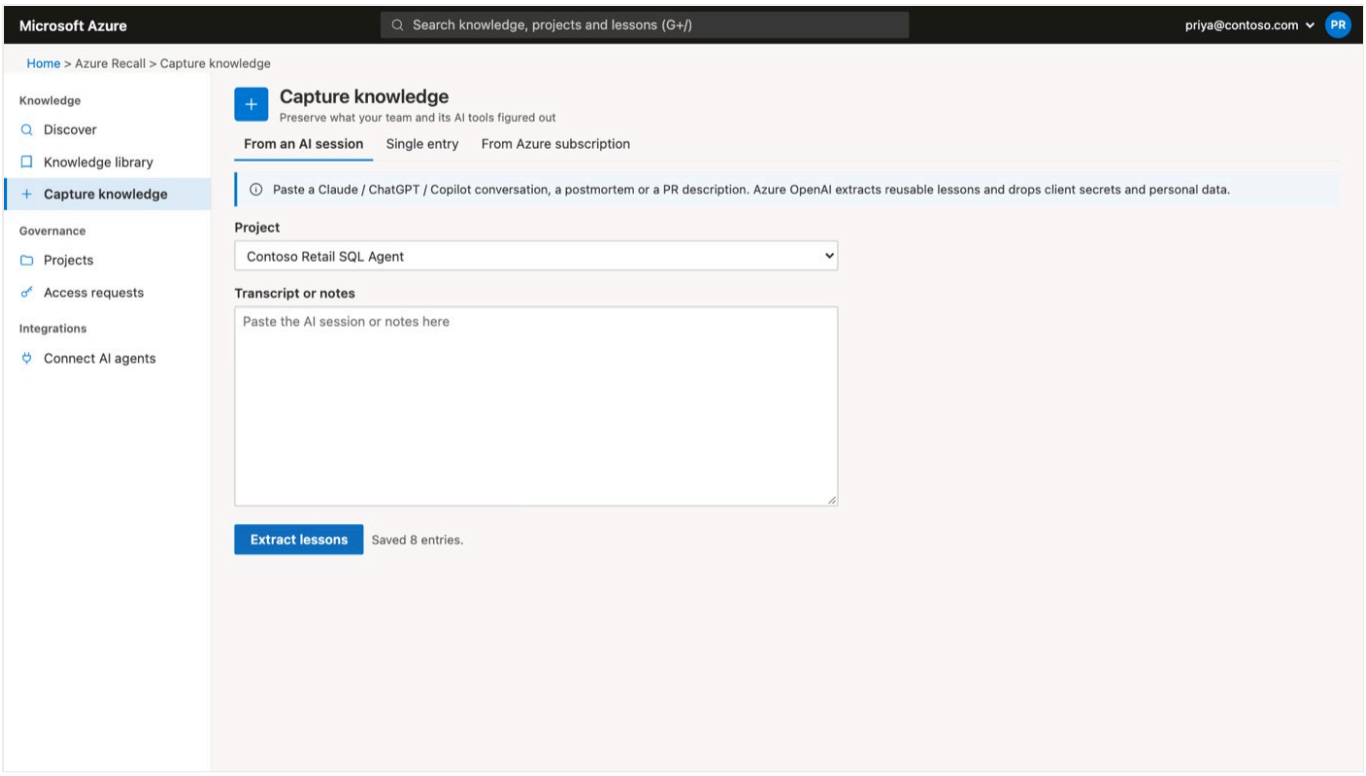
Request access with a business justification, tied to her Entra ID identity.



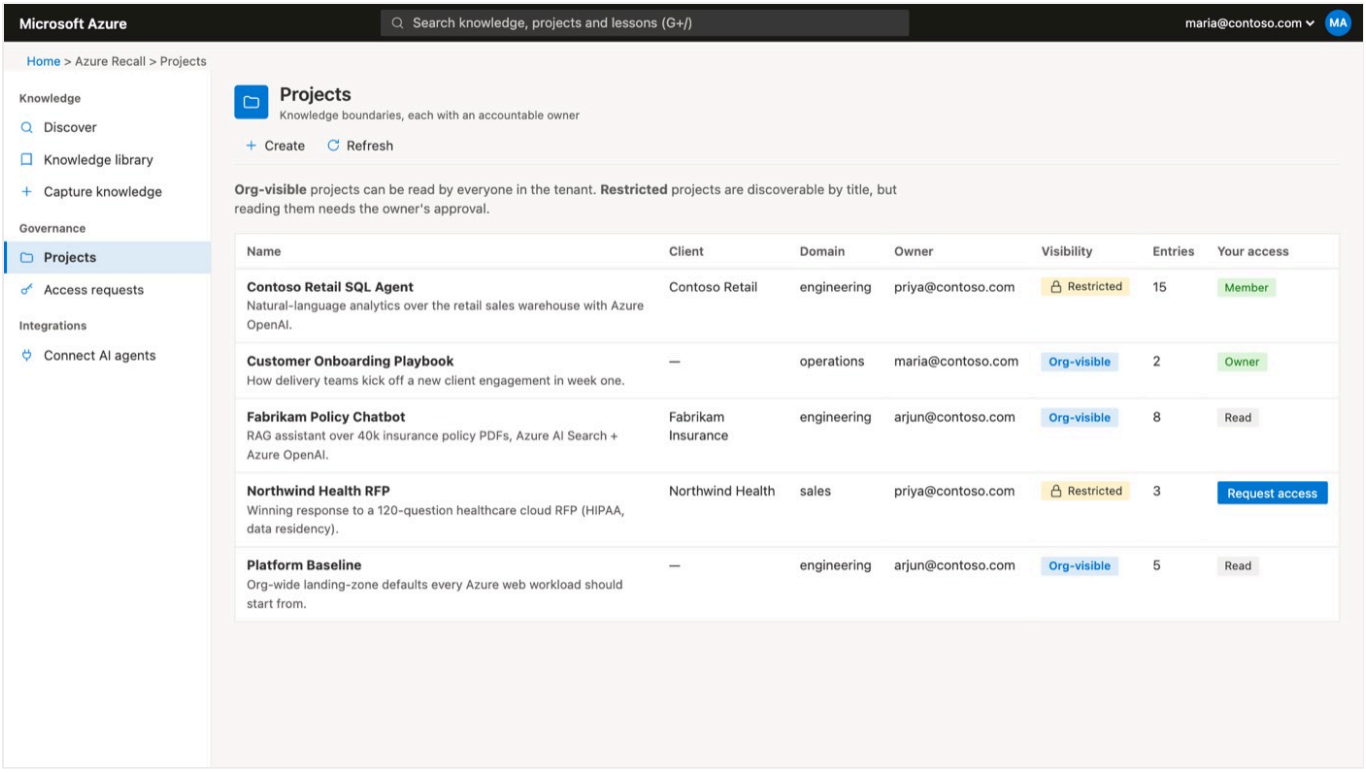
Priya, the project owner, approves or denies in Access requests.



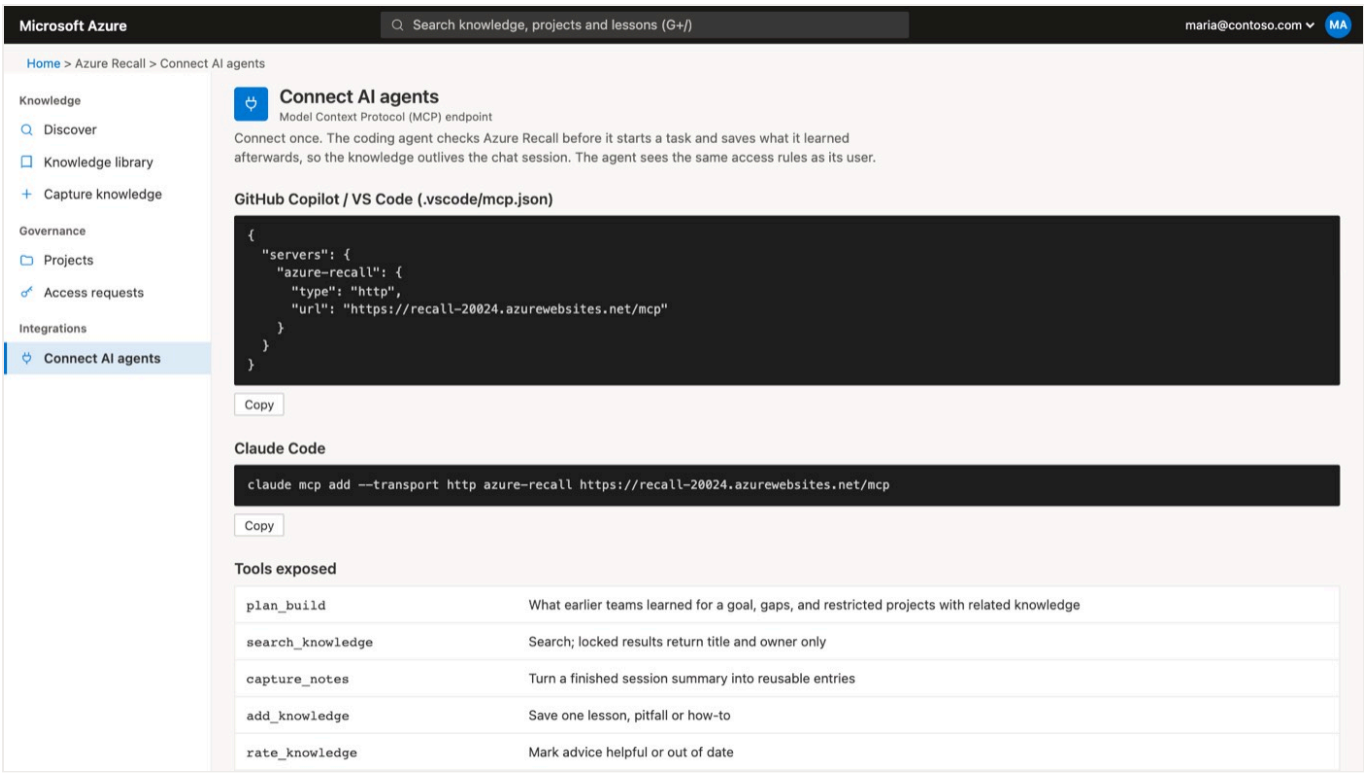
After approval: an Azure OpenAI checklist that cites Contoso's validated lessons.



Capture: a Claude session becomes reusable lessons in the owner's project.



Projects across engineering, sales and operations, each with an owner and visibility.



Connect AI agents over MCP: GitHub Copilot (VS Code) and Claude Code.

10. Business model

Azure Marketplace offer

Azure Application (solution template or managed app). Deploys into the customer's tenant, so knowledge never leaves it: an easy sell to regulated industries.

Azure consumption

App Service, Cosmos DB, Azure OpenAI and, at scale, AI Search grow with every team and every lesson.

Microsoft AI ecosystem

Strengthens GitHub Copilot and Microsoft 365 Copilot with organizational memory; makes Entra ID the control point for knowledge.

Customer value

Less duplicated research, faster delivery on repeat problems, fewer repeated mistakes, knowledge retained when people leave.

11. Roadmap

Item	Value
Azure AI Search	Hybrid vector + semantic ranking with per-project security filters, for knowledge bases beyond ~1,000 entries
Logic Apps / Teams notifications	Owners get approval requests in Teams; time-boxed access grants
Copilot Studio agent	Ask Azure Recall from Microsoft 365 Copilot and Teams
Entra ID groups + Purview labels	Group-based membership; sensitivity labels on entries
Automatic capture	From GitHub pull requests and Azure DevOps work items

Every finished project makes the next one faster. Azure Recall preserves knowledge when people, chats and projects end.